

Problem Statement

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Redshift and blueshift are phenomena wherein light from distant objects is “stretched out”; that is, the observed wavelength increases or decreases relative to the emitted wavelength. This affects the light received on Earth, and can be defined by the following equation:

$$z = \frac{v}{c\sqrt{1 - \frac{v^2}{c^2}}} \quad (1)$$

wherein z is the redshift¹, v is the recessional velocity, in m s^{-1} , and c is the speed of light in a vacuum, approximately $299\,792\,458\,\text{m s}^{-1}$.

This is a problem for cosmologists and spectrographers. Cosmologists need accurate images to study the universe, and whilst tools exist to remedy this, they are nonetheless very slow and inefficient. Spectrographers — especially extraterrestrial spectrographers — rely on spectral lines² to determine the elemental composition of planets’ atmospheres, stars’ atmospheres, and more. Our product aims to enhance our interpretation of redshift³ by accounting for the redshift digitally.

¹If $z = 0$, then there is no redshift or blueshift. If $z < 0$, it is blueshift. And if $z > 0$, then there is redshift.

²which appear in light, i.e., electromagnetic radiation.

³Not stop it, because that would require stopping the expansion of the universe, which is beyond our budget.^[citation needed]